

<https://github.com/shrihanhegdebca25-blip/DBMS-4-6>

```
CREATE TABLE STUDENT (  
    Student_ID INT,  
    Student_Name VARCHAR(50),  
    Class VARCHAR(100),  
    Marks INT,  
    Fee_Paid DECIMAL(10,2),  
    Date_of_Birth DATE,  
    Admission_Date DATE  
);
```

```
INSERT INTO STUDENT  
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth,  
Admission_Date)  
VALUES  
(201, 'Mohith', '10A', 78, 42000.00, '2008-03-14', '2024-06-10'),  
(202, 'Virat Kohli', '10A', 95, 52000.00, '2007-09-22', '2024-06-12'),  
(203, 'Samay Raina', '10B', 69, 46000.00, '2008-01-17', '2024-06-11'),  
(204, 'Cristiano Ronaldo', '10B', 83, 49000.00, '2007-12-05', '2024-06-15'),  
(205, 'Thor', '10B', 72, 43000.00, '2008-05-28', '2024-06-13');
```

4. Implement SQL Functions: Conversion Functions, and Date Functions.

```
SELECT CAST('2026' AS UNSIGNED) + 8 AS Cast_Result;
```

```
SELECT CONVERT('456', SIGNED) + 9 AS Convert_Result;
```

```
SELECT Student_Name, CAST(Fee_Paid AS SIGNED) AS Fee_Integer  
FROM STUDENT;
```

```
SELECT Student_Name, CAST(Marks AS CHAR) AS Marks_Text  
FROM STUDENT;
```

```
SELECT STR_TO_DATE('21-11-2025', '%d-%m-%Y') AS Converted_Date;
```

```
SELECT Student_Name, DATE_FORMAT(Admission_Date, '%d %b %Y') AS  
Admission_Formatted  
FROM STUDENT;
```

```
SELECT Student_Name, FORMAT(Fee_Paid, 2) AS Fee_Formatted
FROM STUDENT;
```

-- Date functions

```
SELECT Student_Name,
       YEAR(Date_of_Birth) AS Birth_Year,
       MONTH(Date_of_Birth) AS Birth_Month,
       DAY(Date_of_Birth) AS Birth_Day,
       DAYNAME(Date_of_Birth) AS Birth_Weekday
FROM STUDENT;
```

```
SELECT Student_Name,
       Admission_Date,
       DATE_ADD(Admission_Date, INTERVAL 2 YEAR) AS Two_Years_Later,
       DATE_SUB(Admission_Date, INTERVAL 20 DAY) AS Twenty_Days_Before
FROM STUDENT;
```

```
SELECT Student_Name,
       DATEDIFF(CURDATE(), Admission_Date) AS Days_Since_Admission,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;
```

```
SELECT CURDATE() AS Today, LAST_DAY('2025-11-10') AS Month_End;
```

5. Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.

```
SELECT 18 + 7 AS Addition, 18 - 7 AS Subtraction, 18 * 7 AS  
Multiplication,  
       18 / 7 AS Division, 18 DIV 7 AS Int_Division, 18 % 7 AS Modulus;
```

```
SELECT Student_Name, Marks,  
       Marks + 7 AS Marks_Plus_7,  
       Marks * 3 AS Marks_Triple,  
       Fee_Paid / 10 AS Monthly_Fee,  
       Marks % 9 AS Marks_Mod_9  
FROM STUDENT;
```

-- Comparison

```
SELECT Student_Name, Marks FROM STUDENT WHERE Marks > 75;  
SELECT Student_Name, Marks FROM STUDENT WHERE Marks < 69;  
SELECT Student_Name, Marks FROM STUDENT WHERE Marks >= 83;
```

```

SELECT Student_Name, Marks FROM STUDENT WHERE Marks <= 72;
SELECT Student_Name, Class FROM STUDENT WHERE Class = '10A';
SELECT Student_Name, Class FROM STUDENT WHERE Class <> '10A';

SELECT 7 = 7 AS Equal, 7 <> 4 AS Not_Equal, 7 > 9 AS Greater,
       7 <=> 7 AS Null_Safe_1, NULL = NULL AS Null_Eq, NULL <=> NULL AS
Null_Safe_2;

```

-- Logical

```

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10A' AND Marks > 75;

```

```

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10B' OR Marks > 90;

```

```

SELECT Student_Name, Marks
FROM STUDENT
WHERE NOT (Marks >= 72);

```

```

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE (Class = '10A') XOR (Marks > 75);

```

6. Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples Special Operators, and Set Operations(UNION)

```

CREATE TABLE Sports_Club (Student_Name VARCHAR(50), Class VARCHAR(100));
CREATE TABLE Music_Club (Student_Name VARCHAR(50), Class VARCHAR(100));

```

```

INSERT INTO Sports_Club VALUES ('Mohith', '10A'), ('Cristiano Ronaldo', '10B'),
('Thor', '10B');
INSERT INTO Music_Club VALUES ('Virat Kohli', '10A'), ('Cristiano Ronaldo', '10B'),
('Samay Raina', '10B');

```

-- Special operators

```

SELECT Student_Name, Marks FROM STUDENT WHERE Marks BETWEEN 70 AND 90;

```

```

SELECT Student_Name, Class FROM STUDENT WHERE Class IN ('10A');

```

```
SELECT Student_Name, Marks FROM STUDENT WHERE Marks NOT IN (69, 95);
```

```
SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE 'M%';
```

```
SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '_a%';
```

```
SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '%a';
```

```
SELECT Student_Name, Fee_Paid FROM STUDENT WHERE Fee_Paid IS NOT NULL;
```

```
SELECT Student_Name, Marks  
FROM STUDENT  
WHERE Marks > ALL (SELECT Marks FROM STUDENT WHERE Class = '10B');
```

```
SELECT Student_Name, Marks  
FROM STUDENT  
WHERE Marks > ANY (SELECT Marks FROM STUDENT WHERE Class = '10B');
```

```
-- Set operations
```

```
SELECT Student_Name, Class FROM Sports_Club  
UNION  
SELECT Student_Name, Class FROM Music_Club;
```

```
SELECT Student_Name, Class FROM Sports_Club  
UNION ALL  
SELECT Student_Name, Class FROM Music_Club;
```